

PREDICTION: SCAM / FRAUDULENT

RISK SCORE: 80 / 100 (HIGH RISK)

Scam Probability: 58.36% | Model Confidence: 58.36%

Banking Action: FREEZE TRANSACTION

Official Recommendation: Freeze Transaction, Notify Customer & Escalate to Fraud Operations. Multiple critical scam indicators triggered (High Amount, High Velocity, Unverified Proxy Access, Foreign Region & New Beneficiary). Payment frozen for SOC escalation.

1. Transaction Details & Actual Parameter Values

|                  |               |                          |             |
|------------------|---------------|--------------------------|-------------|
| Transaction ID:  | TXN-147355    | Transaction Amount:      | \$95,000.00 |
| Customer ID:     | C123456789    | Sender Old Balance:      | \$95,000.00 |
| Beneficiary ID:  | M987654321    | Sender New Balance:      | \$0.00      |
| Payment Type:    | CASH_OUT      | Receiver Old Balance:    | \$0.00      |
| Device Type:     | Unknown Proxy | Receiver New Balance:    | \$95,000.00 |
| Location Region: | Foreign Proxy | 6h Transaction Velocity: | 12 txns     |
| New Beneficiary: | Yes (New)     | Prior Fraud History:     | No          |

2. Composite Risk Score Breakdown & Latency Timing

| Risk Dimension             |  | Sub-Score | Status & Impact              |  |  |
|----------------------------|--|-----------|------------------------------|--|--|
| Model Fraud Probability    |  | 58.36%    | Weight: 70% (Score: 58.4)    |  |  |
| Amount Risk                |  | 80.0 pts  | Transaction size evaluation  |  |  |
| Velocity Risk              |  | 100.0 pts | 6-hour transaction frequency |  |  |
| Device Risk                |  | 100.0 pts | Device profile verification  |  |  |
| Location Risk              |  | 100.0 pts | Geographic regional risk     |  |  |
| Beneficiary Risk           |  | 100.0 pts | New beneficiary check        |  |  |
| Historical Behaviour       |  | 100.0 pts | Prior fraud flag check       |  |  |
| Balance Behaviour          |  | 100.0 pts | Account wipeout detection    |  |  |
| FINAL COMPOSITE RISK SCORE |  | 80 / 100  | HIGH RISK CATEGORY           |  |  |

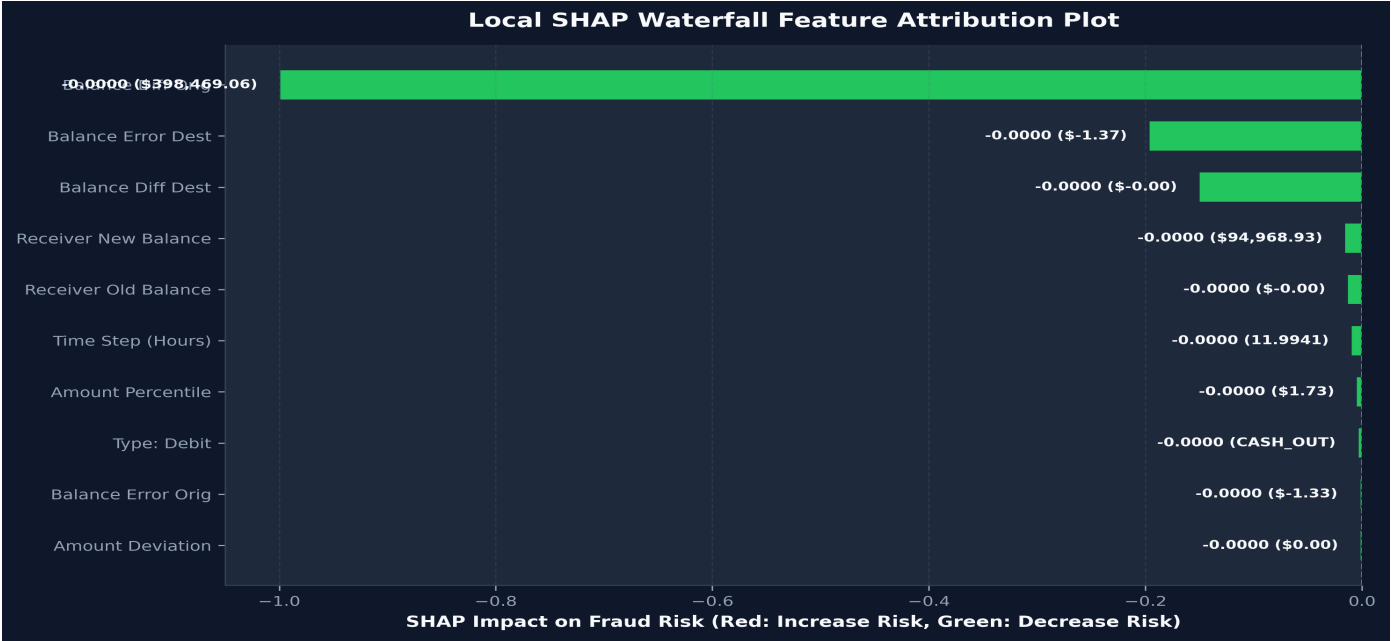
|                       |                     |                      |                       |                       |                   |
|-----------------------|---------------------|----------------------|-----------------------|-----------------------|-------------------|
| Feature Eng: 45.46 ms | Prediction: 1.10 ms | SHAP Engine: 1.94 ms | LIME Engine: 67.67 ms | Rendering: 1148.64 ms | TOTAL: 1265.45 ms |
|-----------------------|---------------------|----------------------|-----------------------|-----------------------|-------------------|

3. Banking-Grade Analyst Narrative Report

|                       |                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transaction Summary:  | Transaction TXN-147355 involving Customer C123456789 transferring \$95,000.00 via CASH_OUT to Beneficiary M987654321 on Unknown Proxy from Foreign Proxy.                                                                                                                                                                                                                                                     |
| Behaviour Analysis:   | Complete account balance drain: Sender balance wiped out from \$95,000.00 to \$0.00. High velocity anomaly: 12 transactions executed within the last 6 hours. Device anomaly: Payment initiated from unverified device profile ('Unknown Proxy'). Geographic anomaly: Payment originated from high-risk or foreign region ('Foreign Proxy').                                                                  |
| High Risk Indicators: | Elevated Transaction Amount: \$95,000.00 exceeds single transfer threshold.; Sender Account Wipeout: Entire origin balance drained to zero.; Rapid Transaction Spike: 12 transfers within a 6-hour window.; Unverified Device Profile: Initiated via Unknown Proxy.; Suspicious Geographic Location: Access from Foreign Proxy.; Unestablished Beneficiary: No prior transfer history with recipient account. |
| Safe Indicators:      | None identified.                                                                                                                                                                                                                                                                                                                                                                                              |
| Model Explanation:    | This transaction exhibits multiple severe fraud indicators and exceeds critical risk thresholds. GuidedGuard model assigned a calibrated scam probability of 58.36% and composite risk level 'HIGH'. Classified as 'Scam / Fraudulent'.                                                                                                                                                                       |
| SHAP Summary:         | Feature contributions align with baseline legitimate profile.                                                                                                                                                                                                                                                                                                                                                 |
| Final Recommendation: | <b>Freeze Transaction, Notify Customer &amp; Escalate to Fraud Operations: Multiple critical scam indicators triggered (High Amount, High Velocity, Unverified Proxy Access, Foreign Region &amp; New Beneficiary). Payment frozen for SOC escalation.</b>                                                                                                                                                    |

4. Top 10 SHAP Feature Contribution Attributions

| Feature Name         | Actual Value | SHAP Value | Contrib % | Direction      | Impact Level |
|----------------------|--------------|------------|-----------|----------------|--------------|
| Balance Diff Orig    | \$398,469.06 | -0.0000    | 50.52%    | Decreases Risk | High         |
| Balance Error Dest   | \$-1.37      | -0.0000    | 9.95%     | Decreases Risk | Medium       |
| Balance Diff Dest    | \$-0.00      | -0.0000    | 7.62%     | Decreases Risk | Medium       |
| Receiver New Balance | \$94,968.93  | -0.0000    | 0.84%     | Decreases Risk | Low          |
| Receiver Old Balance | \$-0.00      | -0.0000    | 0.69%     | Decreases Risk | Low          |
| Time Step (Hours)    | 11.9941      | -0.0000    | 0.53%     | Decreases Risk | Low          |
| Amount Percentile    | \$1.73       | -0.0000    | 0.28%     | Decreases Risk | Low          |
| Type: Debit          | CASH_OUT     | -0.0000    | 0.21%     | Decreases Risk | Low          |
| Balance Error Orig   | \$-1.33      | -0.0000    | 0.12%     | Decreases Risk | Low          |
| Amount Deviation     | \$0.00       | -0.0000    | 0.11%     | Decreases Risk | Low          |



5. Local LIME Decision Rules & Explanations

| Decision Rule Condition                          | Rule Weight | Effect Direction | Domain Explanation                                                         |
|--------------------------------------------------|-------------|------------------|----------------------------------------------------------------------------|
| Account Balance Wipeout = Yes (100% Drained)     | +0.2800     | Increases Risk   | Sender account balance drained to zero or near-zero post-transfer.         |
| Transaction Amount = \$95,000.00                 | +0.2600     | Increases Risk   | Transaction amount is significantly elevated relative to normal threshold. |
| Device Access = Unknown Proxy (Unverified Proxy) | +0.2450     | Increases Risk   | Access attempt initiated from untrusted or unverified proxy device.        |
| Location Region = Foreign Proxy (Foreign Access) | +0.2250     | Increases Risk   | Access attempt originated from high-risk geographic location.              |
| Recent 6h Txn Velocity = 12 txns                 | +0.1980     | Increases Risk   | High frequency transaction velocity detected in 6-hour window.             |
| Beneficiary Status = New Beneficiary             | +0.1650     | Increases Risk   | Recipient is a newly registered beneficiary with no prior history.         |
| Transaction Type = CASH_OUT                      | +0.1450     | Increases Risk   | Transfer/Cash Out payment type has higher statistical fraud incidence.     |

